IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2024

POWER SYSTEM PLANNING

Friday, 3 May 10:00 am

Time allowed: 3 hours

There are THREE questions on this paper.

Answer all questions.

CLOSED BOOK.

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : M. O'Malley

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Answer ALL Questions

1. Question on renewables in power system planning

a) A national power system is optimally planned and has an existing generation mix shown in Table 1 below with a minimum demand of 8.36 GW.

Table 1. Existing generation mix for a national power system.

Technology	Rated Capacity (GW)	Capacity Value	Cost (£/MWh)	GHG Emissions (gCO_2/kWh)
Gas	1	0.96	60	500
Nuclear	12	0.95	10	0

i) What is the expected maximum output of the gas and nuclear generation and the maximum demand that can be served? [4]

ii) Assuming that the demand is uniformly distributed between the minimum and the maximum, calculate the yearly electrical energy production, capacity factors for gas and nuclear, yearly emissions and emissions per kWh generated. [10]

iii) Typically, nuclear generation is inflexible. The power system has a seasonal variation in demand with the demand being on average lower in the summer than the winter by 3GW. Is nuclear generation in this power system flexible or inflexible? Please very briefly justify your answer. [5]

b) The government has set a policy target of net zero without adding to the nuclear fleet. Wind and solar resource characteristics are given in Table 2 below, and it is assumed they are emissions free.

Table 2. Wind and solar resource characteristics

Technology	Capacity Factor (%)	Capacity Value for installed capacity 0 to 1 GW	Capacity Value for installed capacity 1 to 3 GW	Capacity Value for installed capacity greater than 3GW	LCOE (£/MWh)
Wind	40	0.35	0.25	0.15	40
Solar	10	0.05	0.03	0.01	30

i) How much wind will need to be installed to replace the gas generation and maintain reliability? [2]
ii) Assuming flexible nuclear, estimate the wind curtailment. [3]
iii) How much solar will need to be installed to replace the gas generation and maintain reliability? [2]
iv) Assuming flexible nuclear, estimate the solar curtailment. [3]

v) Which is the best option: wind or solar? Please very briefly justify your answer. [4]

2. Question on risk in power system planning

a) Table 3 below contains the objective function value per scenario of a power system planning model under various plans (noted as PF1-PF5, Plan 1, Plan 2). In detail, PF1 is the perfect foresight plan for scenario 1, i.e., it has been obtained assuming only scenario 1 is plausible. PF2 to PF5 are obtained in the same manner. Plans 1 and 2 have been obtained using methods that consider multiple scenarios within a single optimization problem.

Table 3. Objective function value in £

		Plan						
		PF1	**PF2**	**PF3**	**PF4**	**PF5**	**Plan 1**	**Plan 2**
Scenarios	**S1**	5250	5260	5400	5710	6610	5520	5600
	S2	4280	4265	4340	4580	4820	4460	4520
	S3	3660	3475	3150	3195	3330	3180	3300
	S4	3660	3475	2870	2450	2520	2450	2450
	S5	3660	3475	2870	2380	1960	2450	2310

i) Draw the cumulative risk profiles of the two plans (Plan1, Plan2) assuming that the five scenarios are equiprobable. Please show each step for constructing the cumulative risk profiles and your calculations. [5]

ii) Explain if one of the two plans (Plan1, Plan2) dominates the other deterministically or stochastically. [5]

iii) Explain which of the two plans (Plan1, Plan2) you would recommend to a planning agency that minimizes the maximum regret and comment if this criterion helps mitigate the risk of running an expensive power system. [5]

b) For this part of the question assume that only scenarios S1, S3, and S5 are possible, and their total probability is 1. Using equal probabilities (1/3 for each scenario) in a stochastic program, the planning agency obtained plan SP, shown in Table 4 below. The planning agency can hire an expert, who will indicate only one scenario as plausible. The expert is correct 95% of the time.

Table 4. Objective function value in £

		Plans			
		PF1	**PF3**	**PF5**	**SP**
Scenarios	**S1**	5250	5400	6610	5520
	S3	3660	3150	3330	3180
	S5	3660	2870	1960	2450

i) Draw a decision tree that can help the planning agency calculate the expected value of the expert's imperfect information. Please explain how many nodes the tree has, their order, what each node represents and the rationale you followed to estimate the probabilities used. Please note that you do not need to enumerate decision alternatives and add numerical values for consequences. [8]

ii) Using the information from Table 4, can you estimate the expected value of imperfect information, an upper bound or a lower bound of it? If yes, please provide an estimate. If no, please explain what information you are missing. [10]

3. Question on modelling a new technology

According to the IPCC Climate Change 2023 report "Net zero CO2 energy systems entail in addition to a substantial reduction in overall fossil fuel use, … use of carbon capture and storage (CCS) in the remaining fossil fuel systems." Deploying CCS at a power plant or industrial facility generally entails three major steps: capture, transportation, and storage.

a) Post-combustion carbon capture is the primary method used in existing power plants and is illustrated in Figure 1 below.

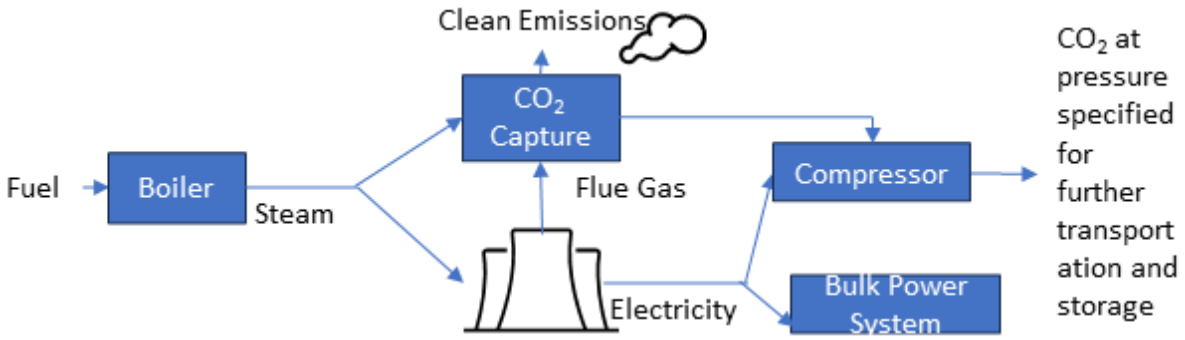


Figure 1. Simplified description of a fossil fuel power plant with carbon capture capabilities. The plant is equipped with (I) capture technology equipment that needs to use steam as input and (II) a compressor that changes the pressure of the CO_2 to delivery standards.

i) List which parameters of the basic capacity expansion model would be different for a fossil fuel power plant with Carbon Capture vs. a fossil fuel power plant without Carbon Capture. Please discuss if you expect their values to be higher, lower, or equal, and why. [8]

ii) Please modify the basic capacity expansion model to consider a carbon tax ($PRICE^{CO2}$) and investments in natural gas power plants with Carbon Capture that can switch on/off the Carbon Capture capability (i.e., their emission rate in $\frac{tnCO_2}{MWh}$ is $\overline{em}$ when the Carbon Capture is off and $\underline{em}$ when the Carbon Capture is on). [12]

b) Assume that there is uncertainty around the carbon tax policy that can be described with two scenarios. In scenario 1, the carbon tax has a constant value (i.e., $PRICE^{CO2}$). In scenario 2, the carbon tax varies per hour of the year ($PRICE_t^{CO2}$) and has the same annual average value as in scenario 1 (i.e., $PRICE^{CO2} = \frac{\sum_t PRICE_t^{CO2}}{\sum_t H_t}$ *where* H_t *is the duration of its interval* t).

i) Explain if you expect CO_2 emissions in scenario 1 to be higher, lower, or equal compared to emissions in scenario 2. [6]

ii) What is the regret of the Expected Value solution in scenario 1? [4]

iii) Show that the equation below holds. [4]

$$EVPI + VSS = \pi_2 \cdot Regret(EVsolution, scenario2)$$

where

- $EVPI$ is the expected value of perfect information.
- VSS is the value of the stochastic solution.
- π_2 is the probability of scenario 2.
- $Regret(EVsolution, scenario2)$ is the regret of the Expected Value solution under scenario 2.

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Addendum

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Basic capacity expansion formulation including investment and operational decisions

$$\min \sum_g (Fixed_Cost_g \cdot cap_g + \sum_t Variable_{Cost_{g,t}} \cdot gen_{g,t}) + \sum_t VOLL \cdot nse_t$$

Decision variables

$$nse_t, cap_g, gen_{g,t}$$

Constraints

$$\sum_g gen_{g,t} + nse_t = Demand_t \quad \forall t$$

$$gen_{g,t} \leq AVAIL_{g,t} \cdot cap_g \ \forall g, t$$

$$nse_t, cap_g, gen_{g,t} \geq 0$$

Source: https://github.com/Power-Systems-Optimization-Course/power-systems-optimization/blob/master/Notebooks/03-Basic-Capacity-Expansion.ipynb

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Two formulations including power flows over existing transmission infrastructure

PTDF-based	B-θ

Objective

$$\min \sum_g (Fixed_Cost_g \cdot cap_g + \sum_t Variable_{Cost_{g,t}} \cdot gen_{g,t}) + \sum_t VOLL \cdot nse_t$$

Decision variables

PTDF-based: $nse_t, cap_g, gen_{g,t}$

B-θ: $nse_t, cap_g, gen_{g,t}, \theta_{n,t}$

Constraints

PTDF-based:

$$\sum_g gen_{g,t} + \sum_n nse_{n,t} = \sum_n Demand_{n,t} \quad \forall t$$

$$P_l^{MIN} \leq \sum_n PTDF_{l,(ref,n)} \cdot \left(Demand_{n,t} - \sum_g df_{g,n} \cdot gen_{g,t} + nse_{n,t}\right) \leq P_l^{MAX} \; \forall l, t$$

$$gen_{g,t} \leq cap_g \; \forall g, t$$

$$nse_t, cap_g, gen_{g,t} \geq 0$$

B-θ:

$$\sum_g gen_{g,t} + \sum_n nse_{n,t} = \sum_n Demand_{n,t} \quad \forall t$$

$$\sum A_{l,n} \cdot B_l \cdot \sum_{n'} A_{l,n'} \cdot \theta_{n',t} = \left(Demand_{n,t} - \sum_g df_{g,n} \cdot gen_{g,t} + nse_{n,t}\right) \; \forall n, t$$

$$P_l^{MIN} \leq \sum_n B_l \cdot A_{l,n} \cdot \theta_{n,t} \leq P_l^{MAX} \; \forall l, t$$

$$gen_{g,t} \leq cap_g \; \forall g, t$$

$$nse_t, cap_g, gen_{g,t} \geq 0$$



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From bilinear term to big M reformulation

Additional Constraints

Bilinear term

$$\sum_n B_l \cdot A_{l,n} \cdot \theta_{n,t} \cdot u_l = flow_{l,t} \quad \forall l_{candidate}$$

$$P_l^{MIN} \cdot u_l \leq flow_{l,t} \leq P_l^{MAX} \cdot u_l \qquad \forall l_{candidate}$$

Additional Constraints

$$-M \cdot (1 - u_l) \leq flow_{l,t} - \sum_n B_l \cdot A_{l,n} \cdot \theta_{n,t} \leq M \cdot (1 - u_l) \quad \forall l_{candidate}$$

$$P_l^{MIN} \cdot u_l \leq flow_{l,t} \leq P_l^{MAX} \cdot u_l \qquad \forall l_{candidate}$$

M: big enough constant

For more details, see section 2.2.3 from book *Investment in Electricity Generation and Transmission Decision Making under Uncertainty by Conejo, Baringo Morales, Kazempour, and Siddiqui.*

Lecture 3

Power Systems Planning Course by Elina Spyrou and Mark O'Malley at Imperial College London. Jupyter Notebook developed by Carlos Matamala.

Data is based on the 'System transformation' scenario within National Grid ESO' Future Energy Scenarios for year 2035

Problem formulation

This yields the following **optimization deterministic problem**:

$$\min \sum_{h \in H} \left(\sum_{g \in G} VarOM_g \times P_{g,h} + \sum_{r \in RES} VarOM_r \times R_{r,h} + VOLL \times NSE_h \right) \tag{1}$$

s.t. (2)

$$\sum_{g \in G} P_{g,h} + \sum_{r \in RES} R_{r,h} + NSE_h = Demand_h \qquad \forall \ h \in H \tag{3}$$

$$0 \leq P_{g,h} \leq CAP_g \qquad \forall \ g \in G, h \in H \tag{4}$$

$$P_{Nuclear,h} = CAP_{Nuclear} \qquad \forall \ h \in H \tag{5}$$

$$0 \leq R_{r,h} \leq CAP_r \times CF_{r,h} \qquad \forall \ r \in RES, h \in H \tag{6}$$

$$NSE_h \geq 0 \qquad \forall \ h \in H \tag{7}$$

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Problem formulation

This yields the following **optimization deterministic problem**:

$$\min \sum_{h \in H} \left(\sum_{g \in G} VarOM_g \times P_{g,h} + \sum_{r \in RES} VarOM_r \times R_{r,h} + VOLL \times NSE_h \right) \quad (1)$$

s.t. (2)

$$\sum_{g \in G} P_{g,h} + \sum_{r \in RES} R_{r,h} + \sum_{s \in STOR} (DISCHARGE_{s,h} - CHARGE_{s,h}) + NSE_h = Demand_h \quad \forall \ h \in H \quad (3)$$

$$0 \leq P_{g,h} \leq CAP_g \quad \forall \ g \in G, h \in H \quad (4)$$

$$P_{Nuclear,h} = CAP_{Nuclear} \quad \forall \ h \in H \quad (5)$$

$$0 \leq R_{r,h} \leq CAP_r \times CF_{r,h} \quad \forall \ r \in RES, h \in H \quad (6)$$

$$0 \leq CHARGE_{s,h} \leq CAP_s \quad \forall \ s \in STOR, h \in H \quad (7)$$

$$0 \leq DISCHARGE_{s,h} \leq CAP_s \quad \forall \ s \in STOR, h \in H \quad (8)$$

$$\sum_{r \in RES} R_{r,h} + (DISCHARGE_{BESS,h} - CHARGE_{BESS,h}) \leq 75\% * Demand_h \quad \forall \ h \in H \quad (9)$$

$$SOC_{s,h} = SOC_{s,h-1} + (CHARGE_{s,h} * Eff_s - \frac{DISCHARGE_{s,h}}{Eff_s}) \quad \forall \ s \in S, h \in H \setminus 1 \quad (10)$$

$$0 \leq SOC_{s,h} \leq SOC_s^{max} \quad \forall \ s \in S, h \in H \quad (11)$$

$$SOC_{s,1} = SOC_{s,t_{end}} = 50\% * SOC_s^{max} \quad \forall \ s \in S \quad (12)$$

$$NSE_h \geq 0 \quad \forall \ h \in H \quad (13)$$